

SNOMED CT Module — Design (Draft)

Module: `hdh.modules.snomed` · **Status:** draft for review · **Date:** 2026-08-11 · **Parent design:** [notes-comprehension-service.md](#) §10

Contributors

Name	Role	Contribution
Ajmal Mahmood	Author / Architect	Requirements, licensing analysis, top-down design

The same principles that built the ICD-10-CM module, applied to a harder ontology: **licensed data** (loader ships, data never does), a **DAG** instead of a tree, **1.6M synonym descriptions** as first-class rows, and **attribute relationships** that carry the clinical semantics interventions need (thrombectomy = *method*: removal + *site*: cerebral artery).

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1. Scope and consumers

In scope (v1): the SNOMED CT **US Edition** — all active concepts (~360k), descriptions (~1.6M), is-a hierarchy, and **defining attribute relationships** (finding-site, method, causative-agent, associated- morphology, using-device/substance, ...). Attributes are promoted to v1 because interventions demand them: "*mechanical thrombectomy of cerebral artery*" is clinically meaningful through its attributes, not its name — and both the neurology specialty pack and care-plan intervention codification consume exactly that.

Consumers: the comprehension service (problems, findings, procedures — its primary `normalize()` target), the care-plan module (SNOMED-coded concerns and interventions per MCC eCare Plan), the agent (`snomed` tools), and the ICD module (curated `maps_to` edges, issue #18).

Out of scope (v1), explicitly: postcoordination (composing new concepts from SNOMED's compositional grammar) — precoordinated concepts only; ICD↔SNOMED map loading (issue #18, separate); non-US editions (structure is identical RF2; a `--edition` flag is future-proofing, not a commitment).

2. Data sources, licensing, and logistics

The load-bearing difference from ICD-10-CM: **CMS files are public domain; SNOMED CT is licensed.** Everything downstream respects that line.

Item	Reality
Source	SNOMED CT US Edition, RF2 format, via NLM's UMLS Terminology Services (UTS)
License	Free UMLS Metathesaurus License — individual signs up at <code>uts.nlm.nih.gov</code> (the US is a SNOMED International member, so US affiliate use is no-cost); approval is days, with an annual usage-report attestation to keep it active.Hdh will never ship with SNOMED CT data
Credential	a personal UTS API key (from the UTS profile page) enables scripted download — <code>HDH_UMLS_API_KEY</code> in <code>.env</code> (gitignored; <code>.env.example</code> documents it, <code>just check-env</code> verifies it)
Cadence	US Edition publishes twice yearly (March 1 / September 1)
Redistribution	prohibited — hdh ships the loader and a synthetic fixture (\$9), never RF2 content; the release zip is cached per-user in <code>~/ .hdh/snomed/<release>/</code> , exactly like the ICD cache but sourced from the user's own credential
Non-US users	member-country affiliates license via their national release center; the loader takes <code>--source <dir></code> so any legitimately obtained RF2 loads

RF2 Snapshot files consumed (v1): `sct2_Concept_Snapshot` (activity, definition status), `sct2_Description_Snapshot` (FSN + synonyms), `sct2_Relationship_Snapshot` (is-a + attributes), and the **US English language refset** (`der2_cRefset_LanguageSnapshot`) to select each concept's preferred term. Full/Delta files are not consumed in v1 (\$5).

3. Schema: what lands where

Same shared tables, new declarations — all via the schema registry, all JSON, per the house pattern:

Piece	Landing	Notes
Concepts	<code>ontology_concepts</code> , <code>ontology='snomed_ct'</code> , <code>code=<SCTID></code> , <code>id='snomed_ct:<SCTID>'</code>	<code>display</code> = US-preferred term; FSN kept in <code>properties.fsn</code>
Kind	<code>kind='concept'</code> (one new enum value, one Alembic migration)	the semantic tag — (<i>disorder</i>), (<i>procedure</i>), (<i>substance</i>)... parsed from the FSN — goes to <code>properties.semantic_tag</code> ; routing logic prefers top-level-hierarchy ancestry over tag string-parsing where both exist
Hierarchy	N <code>parent_of</code> edges (inverted is-a)	path and hierarchy_depth are NULL — DAG; tree conveniences are ICD-private (master doc \$5)

Piece	Landing	Notes
Attributes	<code>edge_type='attribute'</code> (one new enum value) + <code>properties.attribute = {"type_id": ..., "name": "finding_site"}</code>	one generic edge type covers all ~50 attribute types without enum churn; hot attributes get a partial index
Descriptions	NEW entity <code>OntologyTerm</code> (<code>ontology_terms</code>): concept FK, <code>term</code> , <code>term_type</code> (fsn / preferred / synonym), <code>language</code> , <code>active</code>	~1.6M rows; the synonym index the comprehension funnel searches — ICD retro-gains a row per short/long description later
Closure	NEW entity <code>OntologyClosure</code> (<code>ontology_closure</code>): <code>ancestor_id</code> , <code>descendant_id</code> , <code>min_depth</code>	~10M+ rows, composite PK/indexes both directions; private to this module's service — no other module queries it (§8)
Versioning	<code>effective_fy</code> reused as release tag <code>YYYYMM</code> (e.g. 202603); inactive concepts get <code>retired_fy</code> , never deleted	SNOMED <code>effectiveTime</code> preserved in <code>properties</code>

4. First-time load pipeline

The `LoadStage` protocol, reused verbatim — new stages, same contract (fail loudly before `finalize`; ledger row only on full success):

```

1 acquire    UTS download via HDH_UMLS_API_KEY (or --source dir);
              release-zip checksum into the ledger
2 parse      RF2 tab-separated snapshots → typed row streams
              (concepts / descriptions / relationships / langrefset)
3 build      active concepts + preferred-term resolution + semantic
              tags; description rows; edge rows (is-a → parent_of,
              attributes → attribute edges)
4 load       COPY-batched inserts: ~360k concepts, ~1.6M terms,
              ~1.5M+ edges
5 closure    compute is-a transitive closure in memory (array-based
              BFS over ~1.5M edges) → bulk load ~10M closure rows
6 accelerate (PostgreSQL) FTS + trigram GIN over ontology_terms.term;
              partial index on hot attribute types
7 verify     invariants: root concept 138875005 present; every active
              concept has an FSN and a preferred term; closure is
              consistent (acyclicity is implied by closure terminating;
              spot golden concepts §9); counts within tolerance of the
              release's published stats
8 finalize   ledger row: edition, release date, checksums, counts,
              duration

```

Targets (to be benched, not asserted): full US Edition load under 15 minutes on PostgreSQL via `just deps`; closure computation under 5 minutes. SQLite is not a target — this module is where the retirement plan's "PostgreSQL-only for new modules" line is drawn (design icd10cm §5.3 lineage).

5. Periodic updates

Biannual, non-destructive — the ICD fiscal-year playbook:

- v1 strategy: **full snapshot reload per release** (`--force` semantics), with continuity rules: concepts absent-or-inactivated get `retired_fy` set (rows never deleted — `NoteMention` and `maps_to` FKs stay valid); re-activated concepts clear it; description/edge sets are rebuilt per release; closure is recomputed (it is derived data, always safe to rebuild).
- RF2 **Delta** files and historical-association refsets (SAME-AS, REPLACED-BY → edges guiding consumers off retired concepts) are the v2 refinement, worth doing only if reload time or churn tracking proves painful — measured first.
- Each release is a ledger row; `hdh snomed status` shows the active release and retired-concept counts.

6. Graph store: closure and attributes

The two structures that make this module more than a bigger ICD:

Closure answers the questions comprehension and care logic ask constantly — *subsumption*: "is this concept a kind of diabetes / cerebrovascular disorder?" One indexed lookup: `(ancestor, descendant) ∈ closure`. Descendant sweeps ("everything under |Cerebrovascular disease|") are one range scan. No recursive CTE at query time; the CTE runs once per release inside the loader.

Attribute edges carry intervention semantics: thrombectomy's `method → |Removal|`, `procedure site → |Cerebral artery structure|`; a disorder's `finding site` and `causative agent`. v1 loads them all (generic edge type, §3); the *retrieval* API exposes them simply (`attributes(code) → {name: [concepts]}`) and defers clever attribute *querying* (e.g., "all procedures with site under X") until a consumer actually needs it — that query is closure-join-composable when the day comes.

7. Retrieval: the OntologyService implementation

The first full implementation of the master doc's protocol — proving it on the hard case:

Protocol method	SNOMED implementation
<code>lookup(code)</code>	PK fetch + preferred term + semantic tag + FSN
<code>ancestors(code) / descendants(code)</code>	closure joins (SET semantics native)
<code>synonyms(code)</code>	<code>ontology_terms</code> rows
<code>normalize(mention, context)</code>	the ICD funnel shape over the terms index: FTS (progressive relaxation) → trigram fuzz → candidates ranked by term-match quality, semantic-tag fit to the mention type, and ancestor-set context; SapBERT slots here later per master doc §6, bench-gated
extra: <code>subsumes(a, b)</code>	one closure hit — exposed because comprehension's disambiguator and care-gap rules both want it cheap

CLI mirror: `hdh snomed load / status / lookup / search / subsumes / attributes / bench`. Agent tools follow the `icd` pattern (published `build_snomed_tools`), gated on catalog presence.

8. Encapsulation retrofit (master doc §5)

Sequenced **with** this module, not after it: the `OntologyService` protocol lands first (icd10cm retrofitted as implementation #1); `patterns._descend` dispatches per ontology; the quality gate gains the no- `path` -outside-owner check; tree helpers raise `UnsupportedHierarchy` for DAG ontologies; the SQL tool's schema note marks `path` ICD-only. Definition of done for milestone A includes all five items — the silent- empty failure class must be closed *before* the first DAG rows exist.

9. Testing and fixtures — the licensing twist

ICD's fixture was a slice of public-domain data. **SNOMED's cannot be** — a real RF2 extract in a public repo is redistribution. So the committed fixture is **synthetic RF2**: structurally perfect files (real column layouts, valid SCTID check-digits and partition identifiers, coherent is-a/attribute graph) with **fabricated concepts** — a mini clinical world (a disorder tree, a procedure tree with method/site attributes echoing the thrombectomy shape, synonyms, a language refset). Tests prove the loader, closure, DAG semantics, and `normalize()` end-to-end with zero licensed content. Full-release property tests (`@pytest.mark.fullload`) and golden concepts (|Diabetes mellitus|, |Cerebrovascular accident|, a thrombectomy procedure) run only on machines where a licensee has loaded the real edition; CI never touches licensed data. Bench harness identical in shape to `hdh icd bench` with closure-specific patterns (subsumption, descendant sweep).

10. Milestones

Delivers		Proves
A	OntologyService protocol + full §8 retrofit; <code>OntologyTerm</code> + <code>OntologyClosure</code> entities; enum migrations (kind <code>concept</code> , edge <code>attribute</code>) via Alembic	encapsulation closed before DAG rows exist; registry v2 handles the new entities
B	RF2 parser + full pipeline on the synthetic fixture; <code>hdh snomed CLI</code>	DAG load, closure, preferred terms — offline, license-clean
C	UTS download; full US Edition load + closure + bench on <code>just deps</code>	scale reality (360k/1.6M/10M) and honest numbers
D	<code>normalize()</code> funnel over terms + agent tools + <code>subsumes</code>	the comprehension service's primary dependency is ready

After review, the master doc's §10 placeholder collapses to a pointer at this document, and comprehension drill-down resumes at "what is a mention" with its hardest dependency secured.

11. Open questions

1. **Load scope trim?** All 19 top-level hierarchies vs the clinical core (clinical finding, procedure, body structure, substance, pharmaceutical, situation, observable entity) — full is simpler and honest; trimming saves ~30% rows. Lean full; challenge welcome.

2. **ontology_terms** for ICD too, now or later? Retro-loading ICD descriptions as term rows unifies the funnel's search surface before comprehension arrives — probably milestone D, cheap.
3. **Closure storage**: plain table (loader-owned, portable) vs PostgreSQL matview (refresh semantics) — plain table favored; the loader owns derived data explicitly.
4. **Inactive concepts in normalize()**: excluded from candidates, but old notes reference them — include with a **retired** flag and a historical-association pointer (v2)?
5. **Preferred-term language**: US English refset only in v1 — is that acceptable for the synthetic corpus, or does the fixture need to prove multi-language selection works?. V1 US English